

Meaningful Use to Secure Interoperability

Addressing Cybersecurity Risks in Healthcare

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The Health Information Technology for Economic and Clinical Health (HITECH) Act promoted EHR adoption, while the Meaningful Use program provided financial incentives beginning in 2011 to expand EHR use and technology adoption. Meaningful Use encouraged the implementation of technologies such as clinical decision support (CDS), computerized provider order entry (CPOE), electronic prescribing (e-prescribing), and health information exchange (HIE), improving access to health information and contributing to healthcare quality (Centers for Medicare & Medicaid Services [CMS], 2023; Kan, 2011; Jones et al., 2014).

As healthcare organizations connected more systems, the number of access points and dependencies requiring protection increased (Office of the National Coordinator for Health Information Technology [ONC], 2026; Coventry & Branley, 2018). This created competing priorities between interoperability and security. The success of HITECH and Meaningful Use improved healthcare interoperability and access to clinical information, but also changed the nature of cybersecurity risk. The resulting risks cannot be managed uniformly because interoperability creates different security dependencies depending on the technology and clinical workflow. This paper argues that the security consequences of interoperability vary by technology. DICOM creates significant integrity and availability concerns in interconnected imaging environments, while FHIR introduces distinct authorization and confidentiality risks through API-based data exchange.

Digital Imaging and Communications in Medicine (DICOM) presents cybersecurity challenges at the intersection of interoperability, legacy systems, and data protection (National Electrical Manufacturers Association [NEMA], n.d.; Yazdanmehr, 2023a, 2023b). DICOM has expanded remote access to diagnostic imaging through teleradiology, but interoperability with legacy systems creates additional security dependencies (CARE, 2025; NEMA, n.d.). Black Hat

research identified 18.2 million records associated with internet-accessible DICOM servers in the United States, highlighting cybersecurity risks associated with legacy DICOM protocols (Yazdanmehr, 2023a, 2023b). Because DICOM systems interact with PACS and other clinical technologies, a vulnerability in one imaging system can compromise clinical information beyond the affected system, potentially contributing to incorrect diagnoses or treatment decisions. More broadly, ransomware incidents can disrupt clinical systems, delay timely care, and, in severe cases, contribute to patient death (Computer Fraud & Security, 2020).

Fast Healthcare Interoperability Resources (FHIR) presents a different cybersecurity challenge by extending interoperability through API-based communication among EHR platforms, mobile applications, patient portals, and other healthcare technologies (Health Level Seven International [HL7], 2023; Saripalle et al., 2019). This connectivity increases the potential impact of an authorization failure involving sensitive health information. CVE-2026-24487 affected the OpenEMR FHIR CareTeam endpoint and allowed patient-scoped tokens to access care team data belonging to other patients (National Institute for Standards and Technology [NIST], 2026; SentinelOne, 2026). In an interconnected environment, inadequate patient-level authorization can expose information across organizational and technological boundaries. TLS protects FHIR data in transit but does not determine whether an authenticated user is authorized to access a particular patient's data. Therefore, patient-level authorization and role-based access control (RBAC) could have helped prevented the unauthorized access exhibited by CVE-2026-24487, while multi-factor authentication (MFA) strengthens authentication rather than authorization (Vukotich, 2023). The DICOM research revealed that confidentiality controls alone are insufficient when the clinical risk involves image manipulation or service disruptions. Network segmentation and integrity monitoring address different stages of an attack by limiting

lateral movement and detecting unauthorized changes to clinical information (Desjardins et al., 2020; Coventry & Branley, 2018; Kruse et al., 2017; Yazdanmehr, 2023a, 2023b).

The distinction between DICOM and FHIR illustrates that interoperability does not produce a uniform cybersecurity threat. DICOM highlights how interconnected systems can create integrity and availability risks, whereas FHIR illustrates how API-based interoperability can expand authorization and confidentiality risks. Together, the DICOM and FHIR cases demonstrate that interoperability should not be evaluated solely according to whether systems can exchange information. The security implications depend on what is being exchanged, how systems authenticate and authorize access, how clinical data can be modified, and how dependent clinical workflows are on the availability of interconnected systems.

This cybersecurity challenge extends to AI-enabled clinical tools that rely on APIs, cloud services, and third-party applications. Organizations must assess how data, authentication, third-party vendors, and clinical dependencies interact across systems. The NIST CSF 2.0 and AI RMF 1.0 support this risk-based approach (NIST, 2023, 2024). The cybersecurity implications of interoperability are not merely theoretical. Research examining 4,936 U.S. community hospitals found that engagement in health information exchange was associated with an increased probability of an information breach over time, suggesting that expanded information sharing can introduce additional security exposure when organizations and systems become more interconnected (Choi, 2023).

Healthcare leaders and policymakers must apply risk-based practices, including system modernization, vendor oversight, and API governance. Ultimately, interoperability should be measured not only by how much information can be exchanged, but also by whether that information and the systems that deliver it can be trusted when patient care depends on them.

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